

Business Requirements Document – Claims Automation Platform

Executive Summary

Meridian Insurance Group processes over 40,000 property and casualty claims per year through a manual, paper-based workflow. Average claim settlement takes 14 days, driving customer dissatisfaction and high per-claim processing cost. This BRD defines requirements for an AI-driven Claims Automation Platform to reduce settlement time to under 48 hours while maintaining regulatory compliance.

Business Problem

The current process relies on paper intake forms routed by email between five departments: Intake, Underwriting Review, Fraud Investigation, Adjuster Assignment, and Payout. There is no single system of record, causing duplicate data entry, lost documents, and reconciliation errors. Fraud detection is manual and inconsistent, exposing the business to an estimated 3-5% fraudulent-claim leakage.

Functional Requirements

- FR-001: The system must allow customers to submit a claim online with photo/document upload.
- FR-002: The system must automatically extract structured data from uploaded documents using OCR.
- FR-003: The system must run an automated fraud-risk score on every claim at intake.
- FR-004: The system must route claims above a risk threshold to a human fraud investigator.
- FR-005: The system must assign claims below the risk threshold to an available adjuster automatically.
- FR-006: Adjusters must be able to approve, deny, or request more information from a single dashboard.
- FR-007: The system must generate a payout instruction automatically upon adjuster approval.
- FR-008: The system must maintain a complete audit trail of every claim's status changes.

Non-Functional Requirements

- NFR-001: 99.9% uptime SLA.
- NFR-002: p95 API latency under 400ms.
- NFR-003: SOC 2 Type II compliance.
- NFR-004: All PII encrypted at rest and in transit.
- NFR-005: Support 500 concurrent claim submissions at peak (e.g. after a regional weather event).

Proposed Solution

An AI-driven claims automation platform with a React frontend, an API Gateway, a Claims Processing Service, a Fraud Detection Service backed by a machine learning risk model, and a PostgreSQL database. The workflow is: Intake -> OCR Extraction -> Fraud Risk Scoring -> Adjuster Assignment -> Approval -> Payout.

Technology Stack

The solution uses Python FastAPI for the backend, React for the frontend, PostgreSQL for the primary datastore, Redis for caching and queueing, and is deployed on Kubernetes.

Business Benefits

Processing time drops from 14 days to under 48 hours. Operational cost is reduced by 40 percent. Customer satisfaction improves by 25 percent. Fraud leakage is reduced by an estimated 60 percent through automated risk scoring.

Stakeholders

Sponsor: VP of Claims Operations. Business Owner: Director of Claims. Technical Owner: VP of Engineering. Compliance Owner: Chief Compliance Officer.

Conclusion

This platform positions Meridian Insurance Group for scalable, automated, and compliant claims processing, directly addressing the operational and customer-experience gaps in the current manual process.